

# Master's Thesis

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Winter Term 2025/2026

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Philipp Schlatter

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***Numerical Studies of Transition and Turbulence in a Flat-Plate Boundary Layer***

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Institute of Fluid Mechanics (LSTM)  
LSTM Tech. Rep. 01/2025





Title  
of your  
thesis

Master's Thesis, Bachelor's Thesis, Project  
Thesis

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Institute of Fluid Mechanics  
Faculty of Engineering  
Friedrich–Alexander–University Erlangen–Nürnberg  
Prof. Dr. Philipp Schlatter



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| LSTM Technical Report:       | 01/2025                                                |



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# Contents

## List of Figures

# List of Tables

# List of Symbols

semSEM Spectral Element cfdCFD computational fluid dynamics dnsDNS Direct Numerical Simulation  
lesLES Large Eddy Simulation nseNSE Navier-Stokes-Equations ransRANS Reynolds-averaged  
Navier-Stokes daDA Double-Averaged bcB.C Boundary Condition bcsB.Cs Boundary Conditions  
icI.C Initial Condition icsI.Cs Initial Conditions glIGLL Gauss-Lobatto-Legendre [type=]

# A Einleitung

Refer to equations using the following way. In equation (??) we can replace  $G(x)$ . As seen in Fig. ??, however, the referred object always needs to come before the corresponding text. We can refer to literature ? as an important citation (see for instance ?, and others).

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$$G(x) = \sum_0^N \sin^2(x) \tag{A.1}$$

Refer to equations using the following way. In equation (??) we can replace  $G(x)$ . As seen in Fig. ??, we can refer to ? as an important citation (see for instance ?, and others).

## A.1 Important stuff

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**Figure A.1:** The LSTM logo with the latest colours.

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